



RUSSEL FIGO

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SUMMARY

Full Stack Developer with experience in building web and mobile applications, including e-commerce systems, database management platforms, and AI-powered applications. Skilled in developing scalable backend services, designing relational databases, and creating responsive user interfaces. Experienced in integrating machine learning models and building data-driven applications using modern development tools. Passionate about solving real-world problems through efficient system design, clean code practices, and continuous learning of emerging technologies.

EDUCATION

Fresh Graduate of University of Bunda Mulia (GPA 3.69 /4.0) **Aug 2022 - Present**
Major in Informatics. Thesis on "Solana Price Prediction Using Informer-BiLSTM Model".

WORK EXPERIENCE

KPN Corp - FullStack Developer (Intern) **May 2026 - July 2026**

- Vehicle Control System - Web (React · Tailwind CSS · Laravel · WebSocket · MSSQL · Redis · Docker)**
- Built a React-based reporting module with server-side pagination and a client-side XLSX export engine, enabling operators to generate filtered operational audit reports with real-time data synchronization.
 - Engineered a dual-CCTV weighbridge integration with persistent WebSocket-controlled barrier synchronization and a manager override system, enabling real-time gate traffic control for vehicle operations.
 - Implemented CCTV validation protocols and automated RFID lifecycle management, ensuring cards are instantly recycled upon vehicle rejection to eliminate manual intervention at the weighbridge gate.
 - Migrated driver blacklist enforcement from the web dashboard to mobile, decentralizing security checks and empowering ground operators to act on violations without relying on a central system.
 - Refactored vehicle photo storage from Base64 database payloads to a file-path system, significantly reducing database size and improving API response times across photo-heavy endpoints.
 - Redesigned the Vehicle Exit and Dashboard UI with React and Tailwind CSS, optimizing photo rendering via URL endpoints to deliver a faster and more intuitive checkout experience.

- Vehicle Control System - Mobile (Flutter · Dart · REST API)**
- Developed Flutter mobile features for vehicle control operations, including manager override flows for CCTV failures and real-time driver blacklist enforcement for field operators.
 - Built a sequential image-fetching pipeline with auto-retry logic to load CCTV photos efficiently, preventing backend overload while maintaining a responsive user experience.
 - Built an interactive UI with tap-to-zoom image inspection and a navigation drawer, enabling operators to review CCTV evidence and access management modules in a single workflow.
 - Integrated RESTful APIs via Dart's http package to handle secure, asynchronous manager decisions and keep operator workflows synchronized in real time.
 - Improved app stability with graceful error handling, localized state management, and real-time UI feedback, ensuring reliable operation under network disruptions or server failures.

- Warehouse Management System - Web (React · Express.JS · Node.JS · Typescript)**
- Enhanced the interactive warehouse floorplan UI by resolving layout inconsistencies and refining spatial rack logic, improving operator efficiency in locating and managing inventory positions.
 - Designed and implemented end-to-end Goods Issue (GI) and Cancel & Return workflows, including form interfaces and sidebar navigation integration, streamlining warehouse transaction processing.

- Streamlined the Goods Receipt (GR) interface for LOCO shipments by resolving local state synchronization bugs in PO workflows and improving the batch submission experience for warehouse staff.
- Upgraded the Bin Material Panel and Location History tracker, fixing stock quantity display bugs and improving visual feedback to give operators accurate, real-time visibility into inventory movements.

PROJECT EXPERIENCE

Banking Fraud Detection System - Agentic AI + MCP + N8N

March 2026- May 2026

- Developed AI-powered local fraud detection system with multi-agent pipeline, geospatial anomaly detection (Haversine formula), 90-day behavioural analysis, and weighted verdict aggregation.
- Implemented SQL LAG window functions with PARTITION BY and pandas vectorized operations for efficient transaction pair comparison.
- Integrated MCP protocol exposing 9 tools to Claude Desktop for natural language database queries, and N8N for automated fraud monitoring every 5 minutes.
- Built a web dashboard using FastAPI, Chart.js, and Canvas API for visualization of fraud verdicts, sales analytics, and transaction maps across Indonesia.
- Integrated locally hosted Llama3 via Ollama for AI-assisted interpretation of fraud and sales data.

Solana Price Prediction Website

October 2025 - January 2026

- Developed a web-based application for predicting cryptocurrency Solana (SOL) prices using a Informer-Bi-LSTM model.
- Implemented daily price prediction based on user-selected dates and 30-day price forecasting displayed through interactive charts and tables.
- Integrated machine learning model inference into the backend using FastAPI.
- Built data processing pipelines for time series preprocessing, prediction, and result visualization.
- Designed the system to support clear data flow between frontend, backend, and prediction model.

Voice Recognition System for Game – National-Level 1st Place Award Winner

May 2025

- Developed a voice recognition system enabling players to interact with game environments through spoken keywords.
- Designed an end-to-end audio processing pipeline including audio segmentation, MFCC feature extraction, and noise handling.
- Trained and evaluated classification models using TensorFlow, achieving robust keyword detection accuracy.
- Integrated trained models into a real-time game environment for interactive gameplay.

CERTIFICATES

GenAIBIZ - CertNexus

January 2026

- Obtained certification in Generative AI technologies emphasizing business applications and development workflows. Covered concepts such as AI-assisted productivity, responsible AI usage, and practical implementation of generative AI tools.

ADDITIONAL INFORMATION

- **Technical Skills:** Mobile App Development (Flutter), Web Development (React, TypeScript, HTML, CSS, Next.js, Tailwind CSS), Backend Development (FastAPI, Node.js, Express, NestJS, Laravel, REST API), Backend-as-a-Service (Supabase: Auth, Database, Storage), Database Management (PostgreSQL, SQLite, SQL), API Integration, State Management & Application Flow, UI/UX Implementation (Mobile & Web), DevOps & Automation (Docker, n8n), Redis, Version Control (Git), Debugging & Troubleshooting, AI-assisted Development Tools
- **Soft Skills:** Communication, Problem-Solving, Analytical Thinking, Technical Research, Team Collaboration, Adaptability, Time Management, Continuous Learning
- **Languages:** Indonesian, English.